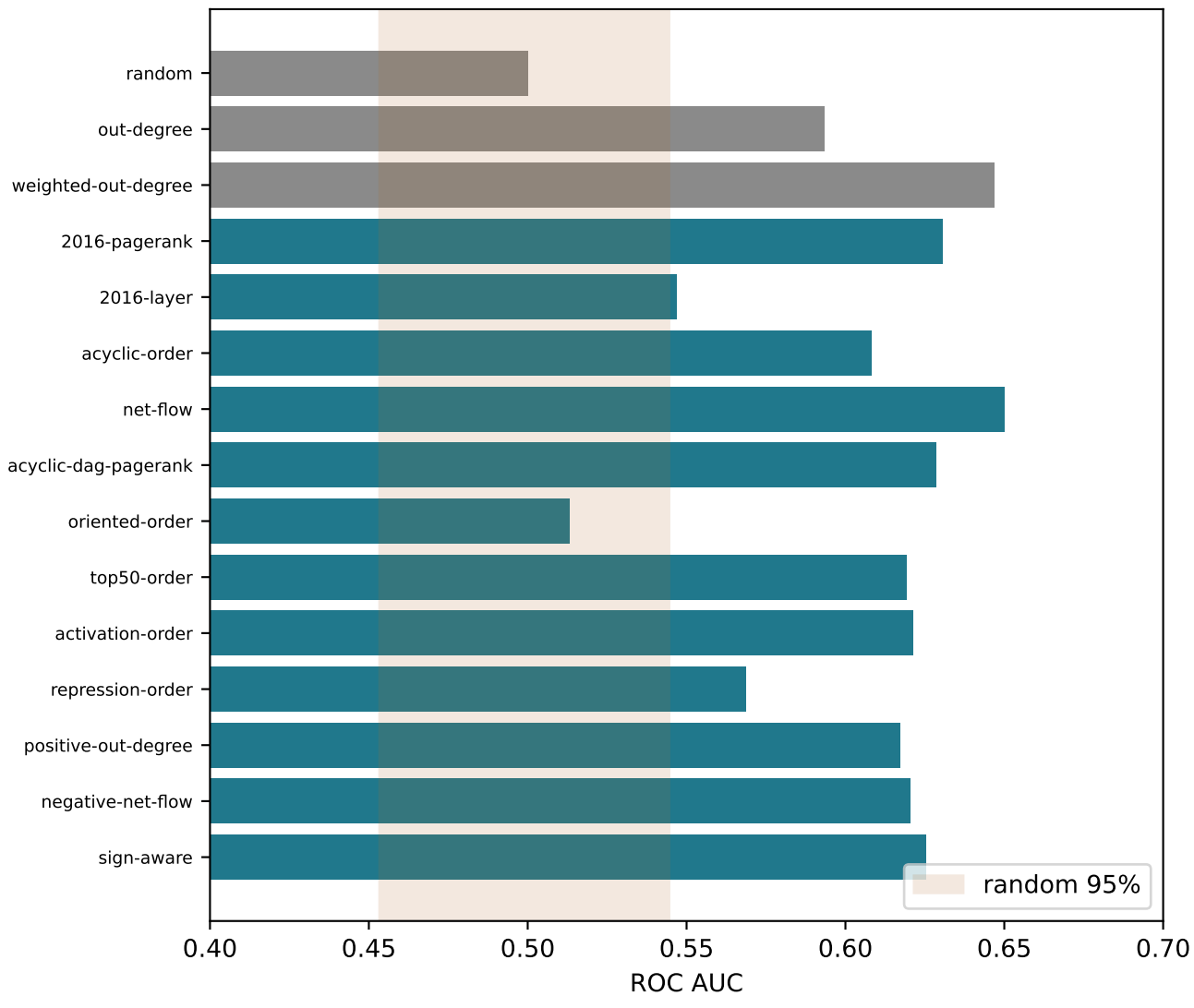
Regulator ranking against reference 'regulators'

2026-11-07 · 740 genes ranked, 200 in regulators.txt · top k = 100

Pipeline	Reference genes in top 100	AUC	p top-k	p AUC	AUC vs best baseline (95%)	p vs baseline
random	27.0 (19–35)	0.500 (0.453–0.545)				
out-degree	39	0.593	0.004	< 0.001	-0.053 (-0.063, -0.045)	< 0.001
weighted-out-degree	46	0.647	< 0.001	< 0.001	+0.000 (+0.000, +0.000)	
2016-pagerank	31	0.631	0.195	< 0.001	-0.016 (-0.052, +0.018)	0.384
2016-layer	74	0.547	< 0.001	0.018	-0.100 (-0.133, -0.065)	< 0.001
acyclic-order	42	0.608	< 0.001	< 0.001	-0.039 (-0.060, -0.015)	0.004
net-flow	42	0.650	< 0.001	< 0.001	+0.003 (-0.006, +0.012)	0.506
acyclic-dag-pagerank	44	0.628	< 0.001	< 0.001	-0.018 (-0.034, -0.002)	0.036
oriented-order	27	0.513	0.548	0.303	-0.134 (-0.162, -0.107)	< 0.001
top50-order	37	0.619	0.012	< 0.001	-0.027 (-0.064, +0.014)	0.182
activation-order	46	0.621	< 0.001	< 0.001	-0.025 (-0.050, +0.001)	0.060
repression-order	32	0.569	0.138	0.001	-0.078 (-0.110, -0.046)	< 0.001
positive-out-degree †	42	0.617	< 0.001	< 0.001	-0.030 (-0.045, -0.016)	< 0.001
negative-net-flow †	31	0.620	0.195	< 0.001	-0.026 (-0.045, -0.009)	0.006
sign-aware †	44	0.625	< 0.001	< 0.001	-0.022 (-0.031, -0.013)	< 0.001

Random: mean and 95% range over random orderings. p: share of random orderings scoring at least as well. Grey bars: random and baselines.
AUC vs best baseline: paired bootstrap over genes (1,000 resamples). †: designed after seeing these results; needs confirming on other data.

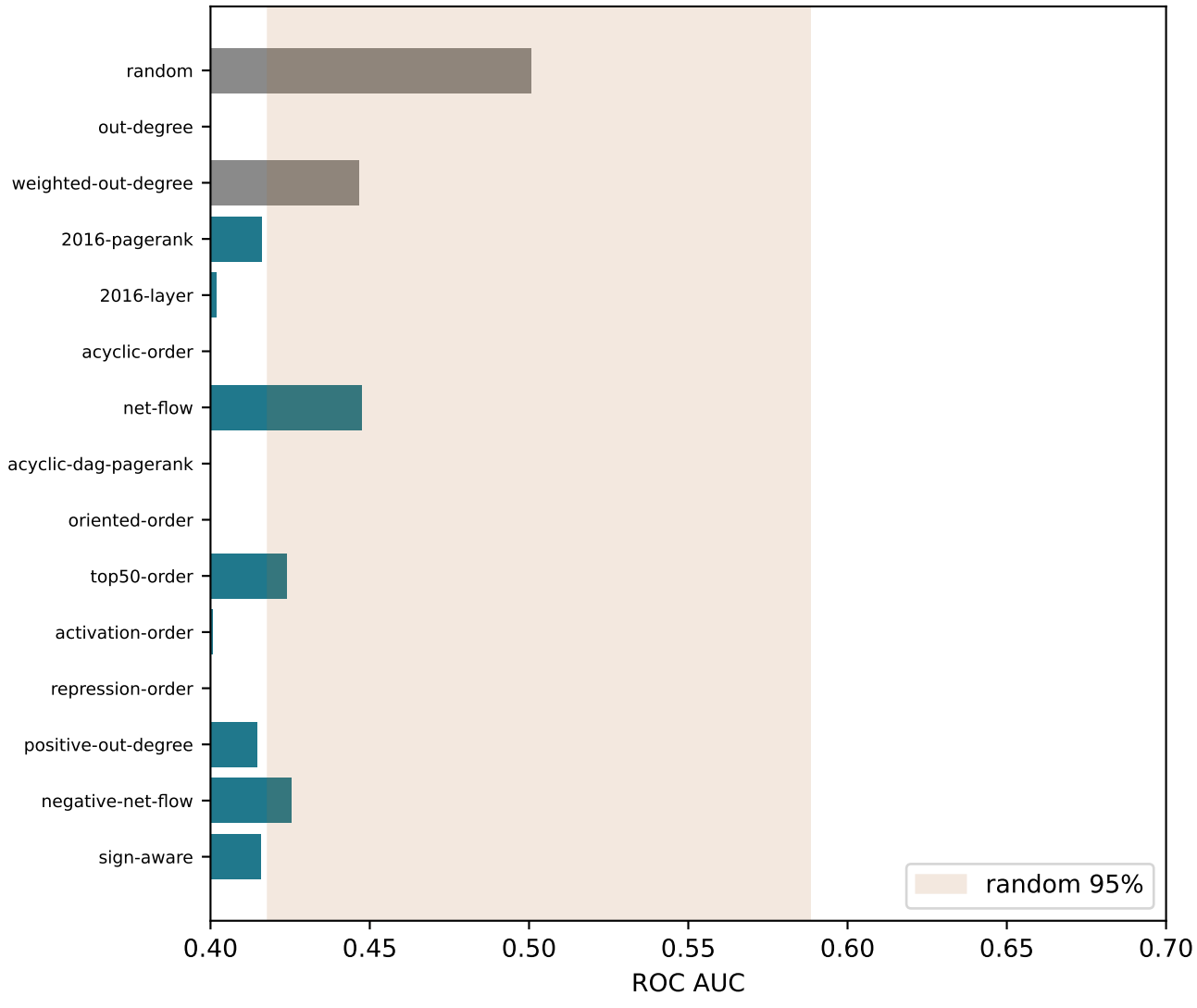


Regulator ranking against reference 'top-layer'

2026-11-07 · 740 genes ranked, 46 in top-regulators.txt · top k = 100

Pipeline	Reference genes in top 100	AUC	p top-k	p AUC	AUC vs best baseline (95%)	p vs baseline
random	6.2 (2–11)	0.501 (0.418–0.588)				
out-degree	3	0.386	0.954	0.997	-0.061 (-0.078, -0.045)	< 0.001
weighted-out-degree	4	0.447	0.884	0.901	+0.000 (+0.000, +0.000)	
2016-pagerank	2	0.416	0.990	0.976	-0.031 (-0.093, +0.030)	0.308
2016-layer	14	0.402	0.001	0.993	-0.045 (-0.114, +0.027)	0.184
acyclic-order	2	0.388	0.990	0.997	-0.058 (-0.104, -0.015)	0.008
net-flow	3	0.447	0.954	0.898	+0.001 (-0.020, +0.022)	0.976
acyclic-dag-pagerank	3	0.395	0.954	0.996	-0.052 (-0.079, -0.028)	< 0.001
oriented-order	1	0.248	0.998	1.000	-0.199 (-0.257, -0.154)	< 0.001
top50-order	3	0.424	0.954	0.967	-0.023 (-0.086, +0.035)	0.410
activation-order	3	0.401	0.954	0.994	-0.046 (-0.094, +0.005)	0.074
repression-order	2	0.383	0.990	0.997	-0.063 (-0.129, -0.003)	0.048
positive-out-degree †	3	0.415	0.954	0.980	-0.032 (-0.056, -0.010)	0.006
negative-net-flow †	0	0.425	1.000	0.965	-0.021 (-0.058, +0.013)	0.212
sign-aware †	3	0.416	0.954	0.976	-0.031 (-0.050, -0.014)	< 0.001

Random: mean and 95% range over random orderings. p: share of random orderings scoring at least as well. Grey bars: random and baselines.
AUC vs best baseline: paired bootstrap over genes (1,000 resamples). †: designed after seeing these results; needs confirming on other data.



Edge checks against curated TF→target pairs

Recall ×: share of curated pairs present as edges, over chance.
Positive/negative ×: share of edges of that sign that are curated pairs, over chance. Sign: agreement with the curated sign (chance).

true

Graph	Pairs	Recall (×)	Positive ×	Negative ×	Sign (chance)
raw graph	2,972	0.849 (25.19×)	30.26×	18.26×	1.00 (0.58)
resolve_reciprocals → drop_sinks → prune_triangles	2,935	0.107 (19.75×)	26.86×	17.33×	1.00 (0.55)
resolve_reciprocals → drop_sinks → prune_triangles → greedy_layering	2,382	0.044 (13.08×)	15.53×	12.47×	1.00 (0.64)
absolute	2,972	0.849 (25.19×)			
absolute → minimum_feedback_order	2,972	0.731 (23.88×)			
orient_reciprocals ratio=2.0 → absolute	2,972	0.532 (20.44×)			
orient_reciprocals ratio=2.0 → absolute → minimum_feedback_order	2,972	0.532 (20.44×)			
top_outgoing k=50 → absolute	2,972	0.764 (37.60×)			
top_outgoing k=50 → absolute → minimum_feedback_order	2,972	0.659 (36.37×)			
keep_sign sign=1 → absolute	2,972	0.589 (30.26×)			
keep_sign sign=1 → absolute → minimum_feedback_order	2,971	0.512 (28.75×)			
keep_sign sign=-1 → absolute	2,972	0.260 (18.26×)			
keep_sign sign=-1 → absolute → minimum_feedback_order	2,972	0.207 (16.43×)			

Pipeline steps

out-degree: Most outgoing edges (raw graph)

score out_degree, descending

Graph: 43,405 edges (19,739 negative), largest strong component 735.

weighted-out-degree: Largest total outgoing weight (raw graph)

score weighted_out_degree, descending

Graph: 43,405 edges (19,739 negative), largest strong component 735.

2016-pagerank: 2016 cleaning, lowest PageRank

resolve_reciprocals() → drop_sinks() → prune_triangles() → score pagerank, ascending

Graph: 7,071 edges (5,184 negative), largest strong component 17.

2016-layer: 2016 cleaning, earliest greedy layer

resolve_reciprocals() → drop_sinks() → prune_triangles() → greedy_layering() → score level, ascending

Graph: 7,071 edges (5,184 negative), largest strong component 17; DAG keeps 3,450 edges.

acyclic-order: Earliest in acyclic order

absolute() → minimum_feedback_order() → score level, ascending

Graph: 43,405 edges (0 negative), largest strong component 735; DAG keeps 38,266 edges.

net-flow: Outgoing minus incoming weight

absolute() → score net_flow, descending

Graph: 43,405 edges (0 negative), largest strong component 735.

acyclic-dag-pagerank: Reverse PageRank on the acyclic DAG

absolute() → minimum_feedback_order() → score dag_reverse_pagerank, descending

Graph: 43,405 edges (0 negative), largest strong component 735; DAG keeps 38,266 edges.

oriented-order: Pairs with a 2x stronger side only, earliest in acyclic order

orient_reciprocals(ratio=2.0) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 33,461 edges (0 negative), largest strong component 1; DAG keeps 33,461 edges.

top50-order: Top 50 outgoing edges per gene, earliest in acyclic order

top_outgoing(k=50) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 27,302 edges (0 negative), largest strong component 734; DAG keeps 23,569 edges.

activation-order: Activating edges only, earliest in acyclic order

keep_sign(sign=1) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 23,666 edges (0 negative), largest strong component 691; DAG keeps 20,799 edges.

repression-order: Repressing edges only, earliest in acyclic order

keep_sign(sign=-1) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 19,739 edges (0 negative), largest strong component 691; DAG keeps 17,455 edges.

positive-out-degree: Most positive outgoing edges

score positive_out_degree, descending

Graph: 43,405 edges (19,739 negative), largest strong component 735.

negative-net-flow: Negative outgoing minus incoming weight

score negative_net_flow, descending

Graph: 43,405 edges (19,739 negative), largest strong component 735.

sign-aware: Rank sum: positive out-degree and negative net flow

score sign_aware, descending

Graph: 43,405 edges (19,739 negative), largest strong component 735.